

Valid research

For Better Health

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Section 1

Designing Clinical Research

What we'll cover today

- What is valid research?
- Types of clinical research
- Bias in clinical research
- Critical appraisal
- AI role

What is Validity?

- Premises
- Conclusion

Types of clinical studies

- Observational studies
- Clinical trials

Cross-sectional studies

- Sample is selected without regard to exposure or disease
- Measurements made at the same time
- Describe variables and their distribution patterns
- For prevalence (pre-test probability)

Case-Control studies

- Participants are selected based on only one OUTCOME
- Retrospective for exposure(s)
- Estimate association (OR and its 95% CI).
- can NOT estimate disease incidence or prevalence

Cohort studies

- Study begins: measure exposure(s) in the assigned cohort
- Follow up
- Study ends: measure outcome(s)

Clinical trials

- Type of trial: parallel, multi-arm, cross-over, factorial, etc
- Unit of allocation: individual, cluster
- Framework: superiority, noninferiority
- Allocation: Non-random, Quasi-random, Random
- Controlled or not
- Phase: Only for drug trials

Bias in clinical studies

- What is bias?
- What do we assess?
- Overarching groups:
 - Selection
 - Information

Bias in clinical studies

- Allocation of intervention bias
- Attrition bias
- Neyman bias
- Misclassification bias
- Temporal ambiguity bias
- Recall bias
- Reporting bias

Critical appraisal

- Critical appraisal addresses
 - Internal validity
 - External validity
 - Statistical validity
 - Clinical relevance
- RoB is nested within critical appraisal.

Critical appraisal

- Appropriate design
- Risk of Bias
- Sound statistical analysis
- Clinically meaningful results
- Generalizable findings

Section 2

Critical Appraisal hands on

Cross-sectional study Comments

- Study label indicates a conceptual confusion.
- Selection bias.
- Temporal ambiguity bias.
- Sample size issues.
- Data analysis errors – AOR 0.37 (0.11 to 0.35)

Case control study Comments

- Is this a case-control design?
- Misclassification bias
- Is the temporal sequence clear?
- No control for Confounding variables
- EEG interpretation workflow concerns
- Sample size issues
- Data analysis issues – check Table 3
- Reference hallucination: reference 12

RCT Comments

- Selection bias
- Serious contradiction in retinopathy inclusion criterion
- Only one HbA1c measurement at the end of follow up
- Intervention was not standardized
- Composite outcome problem
- Follow-up Duration

RCT Comments - cont'd

- Intention-to-Treat Analysis concern
- Sample Size issues
- Statistical analysis issues: HBA1c at 6 months
 - $t = 16.045$ P-value = $2.01e-26$
- Please read the protocol. Any comment?

AI for clinicians

A clinician with strong methodological grounding can identify a plausible but incorrect AI comment

Potential applications

Assist in	Required competency
Literature search	Construct a valid search strategy
Literature screening	Determine the criteria for selecting studies
Data extraction	Inspect, analyze, and interpret Data
Critical appraisal	Assess the risk of bias
Proofreading	Draft a manuscript (English for Academic writing)

Don't

- Use AI to replace critical thinking.
- Rely on AI for conclusions.
- Copy-paste content without understanding it.
- Trust references without verification (AI hallucinates).

Section 3

Thank you